

Institute of Information Technology

Jahangirnagar University, Savar, Dhaka

Subject: Principles of Economics | Tutorial Sheet

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Tutorial Set: Set 1

Total Marks: 30

Chapters: 1, 2 & 4

Section A: Multiple Choice Questions (MCQ)

10 Marks

Choose the best answer for each question. Each MCQ carries 1 mark. Write only the letter.

1. What is scarcity in economics? **[1]**

- a) Having too much money in the economy
- b) When resources are limited relative to unlimited wants
- c) When the government restricts trade
- d) When a market has only one seller

Answer: _____

2. The opportunity cost of going to university is: **[1]**

- a) The tuition fees paid
- b) The cost of textbooks
- c) The best alternative use of time and money given up
- d) The university canteen expenses

Answer: _____

3. According to the Law of Demand, when the price of Hilsa fish rises: **[1]**

- a) Quantity demanded rises
- b) Supply also rises
- c) Quantity demanded falls
- d) The supply curve shifts left

Answer: _____

4. Which of the following causes a SHIFT of the demand curve (not a movement along it)? **[1]**

- a) A change in the price of the good itself
- b) An increase in consumer income
- c) A movement from point A to point B on the same demand curve
- d) None of the above

Answer: _____

5. A point INSIDE the Production Possibilities Frontier (PPF) represents: **[1]**

- a) An efficient use of resources

- b) An impossible level of production
- c) An inefficient use of resources
- d) Maximum possible output

Answer: _____

6. When the price of a good is set BELOW the equilibrium price, the result is: [1]

- a) A surplus (excess supply)
- b) A shortage (excess demand)
- c) No change in quantity demanded or supplied
- d) Market equilibrium

Answer: _____

7. Tea and sugar are examples of: [1]

- a) Substitute goods
- b) Inferior goods
- c) Complementary goods
- d) Normal goods

Answer: _____

8. Which statement is POSITIVE (not normative)? [1]

- a) The government should reduce cigarette taxes
- b) Bangladesh ought to invest more in education
- c) Higher cigarette taxes reduce smoking
- d) Workers deserve a higher minimum wage

Answer: _____

9. If Bangladesh Bank lowers interest rates, employment is likely to rise but inflation may also increase. This illustrates: [1]

- a) Principle 5: Trade makes everyone better off
- b) Principle 10: Short-run trade-off between inflation and unemployment
- c) Principle 6: Markets are usually good
- d) Principle 3: Rational people think at the margin

Answer: _____

10. Given $Q_d = 20 - 3P$ and $Q_s = 10 + 2P$, the equilibrium price P^* is: [1]

- a) $P^* = 1$
- b) $P^* = 3$
- c) $P^* = 2$
- d) $P^* = 5$

Answer: _____

Answer each question briefly and clearly. Each question carries 3 marks.

1. Explain the concept of 'Opportunity Cost' with a relevant Bangladesh example. **[3]**
2. Distinguish between a 'movement along' a demand curve and a 'shift' of the demand curve. Give one example of each. **[3]**
3. What is the Circular-Flow Diagram? Identify its two main groups and two main markets. **[3]**
4. State the Law of Supply. Why does the supply curve slope upward? Use the garment industry as an example. **[3]**

Section C: Long Answer Questions

8 Marks

Answer in detail. Show all workings where calculations are required.

1. The following demand and supply equations describe the rice market in a local bazaar: **[8]**
 $Q_d = 50 - 4P$ $Q_s = 10 + 6P$
 - (a) Find the equilibrium price (P^*) and equilibrium quantity (Q^*).
 - (b) At a price of $P = 5$, calculate the surplus or shortage. State which one it is.
 - (c) Suppose consumer income rises significantly (rice is a normal good). Using the 3-Step Method, analyse the new equilibrium. What happens to price and quantity?
 - (d) Briefly explain Adam Smith's 'Invisible Hand' concept and connect it to your answer above.

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Tutorial Set: Set 2

Total Marks: 30

Chapters: 1, 2 & 4

Section A: Multiple Choice Questions (MCQ)

10 Marks

Choose the best answer for each question. Each MCQ carries 1 mark. Write only the letter.

1. Which of the following is an example of a NEGATIVE incentive? [1]

- a) A subsidy given to farmers for buying fertilizer
- b) A cash prize for top exam scorers
- c) A fine for breaking traffic laws
- d) A bonus for workers who exceed production targets

Answer: _____

2. Bangladesh specializes in garments and trades with Japan for electronics. This is an example of: [1]

- a) Principle 9 — Prices rise when money is printed
- b) Principle 5 — Trade can make everyone better off
- c) Principle 7 — Government can improve markets
- d) Principle 1 — People face trade-offs

Answer: _____

3. Microeconomics is best described as: [1]

- a) The study of inflation and unemployment across the national economy
- b) The study of GDP and money supply
- c) The study of individual households, firms, and specific markets
- d) The study of international trade policy

Answer: _____

4. When both demand and supply increase simultaneously, what is CERTAIN? [1]

- a) Price definitely rises
- b) Equilibrium quantity definitely rises
- c) Price definitely falls
- d) Equilibrium quantity is ambiguous

Answer: _____

5. Harry's income increases, but he buys fewer bus rides. Bus rides are: [1]

- a) Normal goods
- b) Complementary goods

- c) Inferior goods
- d) Substitute goods

Answer: _____

6. The PPF shifts OUTWARD when: **[1]**

- a) Resources are wasted
- b) The economy enters a recession
- c) New technology improves production efficiency
- d) The government increases taxes

Answer: _____

7. Price Elasticity of Demand ($|E_p|$) = 1.8. This means demand is: **[1]**

- a) Perfectly inelastic
- b) Inelastic
- c) Elastic
- d) Perfectly elastic

Answer: _____

8. A garment factory polluting the Buriganga River is an example of: **[1]**

- a) Positive externality
- b) Invisible hand
- c) Negative externality
- d) Market equilibrium

Answer: _____

9. In the Circular Flow Diagram, households sell to firms in the: **[1]**

- a) Market for goods and services
- b) Market for factors of production
- c) Government sector
- d) Foreign exchange market

Answer: _____

10. Which of the following correctly describes marginal thinking (Principle 3)? **[1]**

- a) Always look at total cost vs. total benefit before deciding
- b) Compare the extra benefit of one more unit with the extra cost of that unit
- c) Make large changes to maximize output
- d) Avoid all decisions that have a cost

Answer: _____

Section B: Short Answer Questions

12 Marks

Answer each question briefly and clearly. Each question carries 3 marks.

1. What are 'Determinants of Demand'? List FIVE factors that shift the demand curve, with one example each. **[3]**
2. Explain the concept of 'Market Failure' and the role of government. Use a Bangladesh example involving externalities. **[3]**
3. Distinguish between Positive Economics and Normative Economics. Give two examples of each. **[3]**
4. What is Price Elasticity of Demand? Given $P_1 = 10$, $Q_1 = 200$ and $P_2 = 8$, $Q_2 = 250$, calculate the elasticity and classify it. **[3]**

Section C: Long Answer Question

8 Marks

Answer in detail. Diagram is required where asked. Show all workings.

1. Consider the following scenario for the Dhaka vegetable market: **[8]**
 - (a) Using a well-labelled supply-and-demand diagram, show what happens when a flood destroys vegetable crops in Sylhet (reducing supply), while the population of Dhaka increases (increasing demand). Label the old and new equilibrium points.
 - (b) State the '3-Step Method' and apply it to the scenario above. What happens to price? What happens to quantity? Explain any ambiguity.
 - (c) The government then announces a price ceiling (maximum price) BELOW the new equilibrium. Explain whether a surplus or shortage will result and why.
 - (d) Which of Mankiw's 10 Principles does this government intervention most relate to? Justify your answer.

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Tutorial Set: Set 3

Total Marks: 30

Chapters: 1, 2 & 4

Section A: Multiple Choice Questions (MCQ)

5 Marks

Choose the best answer. Each MCQ carries 1 mark.

1. When the price of Rohu fish falls, demand for Hilsa fish falls too. This means Rohu and Hilsa are: **[1]**

- a) Complementary goods
- b) Inferior goods
- c) Substitute goods
- d) Normal goods

Answer: _____

2. A country's standard of living (Principle 8) most directly depends on: **[1]**

- a) The size of its military spending
- b) Its ability to produce goods and services (productivity)
- c) The total number of people in the country
- d) The amount of gold in its central bank

Answer: _____

3. In a competitive market, who has control over the price? **[1]**

- a) The largest firm in the market
- b) The government
- c) No single buyer or seller — everyone is a price taker
- d) Households only

Answer: _____

4. The scientific method in economics relies on: **[1]**

- a) Lab experiments with full control over all variables
- b) Observation, theory building, and testing with data (often natural experiments)
- c) Government surveys only
- d) Mathematical proofs with no real-world data

Answer: _____

5. If $Q_s = 10 + 2P$ and $P^* = 2$, then equilibrium quantity $Q^* =$ **[1]**

- a) $Q^* = 10$
- b) $Q^* = 14$

c) $Q^* = 12$

d) $Q^* = 16$

Answer: _____

Section B: Short Answer Questions

10 Marks

Answer each question clearly. Marks are as indicated.

1. Define 'Equilibrium Price' and 'Equilibrium Quantity'. Explain what happens in a market when the price is above equilibrium. Use the Eid cattle market in Bangladesh as your example. **[3]**
2. Explain the PPF with reference to its four key concepts: Scarcity, Efficiency, Trade-offs, and Opportunity Cost. **[3]**
3. Explain two reasons why economists sometimes disagree with each other, giving one example for each reason. **[2]**
4. List the three determinants of Supply (factors that shift the supply curve) not related to price. Give a Bangladesh garment industry example for each. **[2]**

Section C: Application / Analytical Questions

15 Marks

Read each scenario carefully and apply economic principles. Show all reasoning clearly.

1. Scenario — The Fertilizer Subsidy: **[6]**

The Bangladesh government announces a Tk. 5 per litre subsidy on diesel for all paddy farmers in the country. Using at least THREE of Mankiw's 10 Principles of Economics, analyse the likely economic effects of this policy. For each principle you use, clearly state the principle, its number, and how it applies to this scenario.

2. Numerical Problem — Ice Cream Market: [6]

Given: $Q_d = 30 - 5P$ **and** $Q_s = -5 + 5P$

- (a) Solve algebraically for equilibrium price P^* and equilibrium quantity Q^* . [2]
- (b) At $P = 4$, calculate whether there is a surplus or shortage and by how much. [2]
- (c) The price of sugar (an input) rises sharply. Using a diagram, show the new equilibrium. [2]

3. Critical Thinking — Bangladesh Bank's Dilemma: [3]

Bangladesh Bank is facing both high inflation (9%) and rising unemployment (7%). Using Principle 10 (short-run trade-off between inflation and unemployment) and the Phillips Curve concept, explain the dilemma the bank faces. What are its two policy options and what are the consequences of each?